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July 15, 1975
SUPERSEDING
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September 5, 1967

FEDERAL SPECIFICATION

WAXING AND CLEANING COMPOUND, IRONER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a water emulsion of laundry wax, which is used to wax flatwork ironers and equipment, and for improving the finish of linens. In addition, this emulsion serves to keep the flatwork ironers clean and free from wax buildup, and to reduce static.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).
PPP-P-1655 - Pail, Plastic, Shipping and Storage.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC; Atlanta, Chicago, Kansas City, MO; Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

D 93 - Flash Point by Pensky-Martens Closed Tester.
D 95 - Water in Petroleum Products and Bituminous Materials by Distillation.
D 127 - Drop Melting Point of Petroleum Wax, Including Petrolatum.
D 217 - Cone Penetration of Lubricating Grease.
D 2834 - Nonvolatile Matter (Total Solids) in Water-Emulsion Floor Polishes, Solvent-Based Floor Polishes, and Polymer-Emulsion Floor Polishes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The material covered by this specification shall be a creamy, lump free, water-emulsion paste which will decrease friction on rolls, chests, and aprons of flatwork ironer and equipment. It shall be free from tackiness and must withstand heat and pressure encountered in flatwork ironers without gumming. When used as directed, it shall keep the ironer chest clean and polished, and reduce static. The material shall comply with the requirements of table I.

TABLE I. Material characteristics

Characteristic	Minimum	Maximum	Test paragraph
Flash point	400°F.	--	4.3.2
Moisture	40.5 percent	41.5 percent	4.3.1
Wax content (nonvolatile)	43 percent	45 percent	4.3.3
Melting point	150°F.	155°F.	4.3.4
Saponification number	60	95	4.3.5
Acid number	6	10	4.3.6
Ash content	0	0.15 percent	4.3.7
Solvent content	14 percent	17 percent	4.3.8
Penetration	250	350	4.3.9
Heat stability	as described		4.3.10
Storage stability	as described		4.3.11

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspections.

4.2.1 Quality conformance inspection sampling. Sampling for inspection and tests shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated. For purposes of sampling for inspection, a lot shall consist of all material of the same type submitted for inspection and delivery at one time. The AQL shall be 2.5 defects per hundred units, inspection level I.

4.2.2 Inspection of materials. In accordance with 4.1, the supplier is responsible for insuring that the materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3 Test methods.

4.3.1 Water content. Water in the sample shall be determined in accordance with ASTM D 95.

4.3.2 Flash point. The flash point shall be determined in accordance with ASTM D 93.

4.3.3 Wax content, nonvolatile. The nonvolatile waxes in the sample shall be determined in accordance with ASTM D 2834.

4.3.4 Melting point. The melting point of the wax shall be determined on the nonvolatile residue left after 72 hours heating at 42°C. ± 2°C., in accordance with ASTM D 127.

4.3.5 Saponification number. The saponification number shall be determined on the nonvolatile residue (see 4.3.4) in the following manner: Weigh out a 5 gram (g.) sample into a 250 milliliter (ml.) flask and add 50 ml. of standard 0.5 normal (N) alcoholic sodium hydroxide (NaOH). Prepare a blank, and reflux both flasks for 75 minutes. Titrate while still hot using standard 0.5 N hydrochloric (HCl) acid, with phenolpthalein indicator, and calculate as follows:

$$\frac{\text{ml. HCl for blank} - \text{ml. HCl for sample} \times 28.05}{\text{g. sample}} = \text{Saponification number}$$

4.3.6 Acid number. The acid number shall be determined on the nonvolatile residue (see 4.3.4) in the following manner: Weigh out a 10 g. sample, dissolve it in a 100 ml. of a 50-50 alcohol-benzene mixture and heat the solution on a water bath to 55°C. Titrate at this temperature with standard 0.1 N NaOH using phenolpthalein indicator. Calculate as follows:

$$\frac{\text{ml. of 0.1 N NaOH} \times 0.0282 \times 100}{\text{g. sample}} = \text{Percent free oleic acid}$$

Percent free oleic acid X 1.99 = Acid number

4.3.7 Ash content. The ash content shall be determined on the nonvolatile residue (see 4.3.4) in the following manner: Weigh out 5 g. into a porcelain crucible and heat gently, allowing the wax to burn off. Finally, the crucible is ignited at a dull red heat until all the carbon is burned off. Cool in a dessicator and weigh. Calculate as follows:

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$$\frac{\text{g. residue}}{\text{g. sample}} \times 100 = \text{Percent ash}$$

4.3.8 Solvent content. The solvent content shall be determined by difference: 100 percent - (percent moisture + percent total solids) = percent solvent.

4.3.9 Penetration. The penetration shall be determined in accordance with ASTM D 217, for a worked sample.

4.3.10 Heat stability. Heat stability shall be determined in the following manner: Two samples of about 3 g. each are weighed into a suitable dish or crucible, and heated in an oven at 105 to 110°C. for 4 hours. At the end of that time, one sample is set aside, and the other is heated at 150°C. for 4 hours. Compare the samples by feeling and rubbing the material between the fingers. There should be no sign of tack or loss of slip in the heated sample, and only slight discoloration.

4.3.11 Storage stability. A 10 to 15 g. sample shall withstand three cycles of 1 hour at 0°F. (-18°C.) and 1 hour at room temperature 77°F. (25°C.), with no physical change. The material shall remain in paste form without separation at temperatures up to and including 110°F. (43°C.). From 115 to 140°F. (45 to 60°C.), it may liquefy, but shall not separate, and on standing at room temperature afterwards, shall revert to the paste form without separation.

4.3.11.1 Duration of storage stability. The material shall be storable for a minimum of 1 year in original unopened container without going rancid or otherwise becoming unsatisfactory for use. In the event of spoilage, the year shall be counted from the date of acceptance of the shipment in question.

4.4 Examination of preparation for delivery. An examination shall be made to determine whether the packing and marking comply with the requirements in section 5 of this specification. Defects should be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. Sampling shall be in accordance with MIL-STD-105, inspection level S-2, and the AQL shall be 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Container (unit and shipping)	Not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flap or distortion of container.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Forty pounds. Forty pounds of compound, as specified (see 6.2), shall be packed in a pail conforming to PPP-P-704, type II or III. Each pail shall be provided with a polyethylene bag to enclose the compound. The top of the bag shall be long enough to permit securing it with ties, clips, or by sealing. Alternatively, 40 pounds of compound shall be packed in a pail conforming to PPP-P-1655, type I, and a polyethylene bag is not required.

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5.1.2 Level B.

5.1.2.1 Forty pounds. Forty pounds of compound, as specified (see 6.2), shall be packed in a pail conforming to PPP-P-704, type II or III. Commercial coatings of the pails will be acceptable. Each pail shall be provided with a polyethylene bag to enclose the compound. The top of the bag shall be long enough to permit securing it with ties, clips, or by sealing. Alternatively, 40 pounds of compound shall be packed in a pail conforming to PPP-P-1655, type I, and a polyethylene bag is not required.

5.1.3 Level C. The compound, in the quantity specified (see 6.2), shall be packed in containers to assure carrier acceptance and safe delivery at destination. The containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.2 Marking.

5.2.1 General. Each container shall be marked with the directions for use.

5.2.2 Civil agencies. In addition to the markings required by 5.2.1 and any special markings required by the contract or order, each shipping container shall be marked in accordance with Fed. Std. No. 123.

5.2.3 Military agencies. In addition to the markings required by 5.2.1 and any special markings required by the contract or order, each shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The laundry wax emulsion covered by this specification is intended as an ironing wax to allow easy passage of goods through the ironer, and as a detergent to keep the ironers clean and polished (see 1.1).

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Special markings required (see 5.2).

MILITARY INTEREST:	CIVIL AGENCY COORDINATING ACTIVITIES:
The Department of Defense has waived coordination of this document until further notice.	GSA-FSS VA-DMS CPSC

PREPARING ACTIVITY: GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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